No.	Status	Date
HIS-UT-002 Rev 1	Issued	2004-03-08

Ultrasonic Thickness Measurement Procedure

1.0 Scope and Application

This procedure establishes the requirements for ultrasonic thickness measurement of piping and pressure components at the Hamilton Plant. It is submitted by Halward Inspection Services Ltd. for formal acceptance by the QSCP Hamilton Inspection Department to ensure compliance with site-specific NDE requirements. The methodology defined herein is intended to support the pre-startup and in-service NDE activities originally outlined in QSCP-NDT-001 Rev 0.

The techniques described are applicable to carbon steel and alloy components where manual pulse-echo ultrasonic methods are required to determine wall loss. All examination activities shall be performed in accordance with the technical principles of ASTM E797 to ensure accuracy and repeatability of data across all monitored locations.

2.0 Personnel Qualifications

All personnel performing ultrasonic thickness measurements under this procedure shall be qualified and certified in accordance with ASNT SNT-TC-1A. Only individuals certified to at least Level I may perform the data collection, provided they are under the direct supervision of a Level II or Level III technician. The interpretation of results and final reporting must be performed by a Level II or higher.

Anil Rao, acting as the Level III for Halward Inspection Services Ltd., is responsible for the technical oversight of this procedure and the certification of the personnel involved. Anil Rao shall ensure that all examiners demonstrate proficiency in the specific equipment and calibration techniques required for the Hamilton Plant assets.

3.0 Equipment and Calibration

Ultrasonic instruments used shall be of the pulse-echo type, capable of digital thickness readout and, where possible, A-scan verification. Calibration shall be performed on materials of the same acoustic velocity as the component being inspected, following the guidelines in ASTM E797. Calibration checks must be performed at the start of each work shift and whenever a probe or battery is changed.

Couplants must be selected based on the surface temperature and finish of the component. For high-temperature applications, specialized couplants and delay-line transducers may be required to prevent signal drift. All calibration data, including the thickness of the reference blocks used, must be recorded on the inspection report for traceability.

4.0 Data Evaluation and Site Rules

The retirement thickness for any component is defined as the line t-min. Any reading obtained that is below the specified t-min must be immediately raised as an NCR. The corrosion rate for the component shall be determined by selecting the larger of the long-term rate, calculated since the first reading, or the short-term rate, calculated over the latest pair of readings that are at least 3 years apart.

Remaining life is calculated by dividing the difference between the current thickness and t-min by the determined corrosion rate. The next inspection interval shall be set at no more than half the remaining life, subject to the class maximums. For Class 1 systems such as sour gas or regenerator overheads, the maximum interval is 5 years. Class 2 systems like amine or treated gas are capped at 10 years, and Class 3 systems like cooling water are also capped at 10 years.

5.0 Reporting and Documentation

All inspection findings shall be documented on reports that reference QSCP-NDT-001 Rev 0 to maintain consistency with the ATU-100 pre-startup data. Reports must include the component identifier, the measured thickness, the calculated corrosion rate, and the projected remaining life. Any areas showing significant localized thinning must be mapped to determine the extent of the degradation.

Final reports shall be reviewed for accuracy and compliance with this procedure. The documentation must clearly state the next due date for both thickness measurements and external visual inspections based on the class maxima: 5 years for Class 1 and Class 2, and 10 years for Class 3. All records shall be maintained as part of the permanent asset integrity file for the Hamilton Plant.

Sign-off	Name	Function	Date	Signature
Prepared	Anil Rao	Level III, SNT-TC-1A UT/RT Level III	2004-03-08	_____